

AI 553 Speech and Audio Processing
Assignment - 1

(11) (a) $x_1(t) = j e^{j10t}$

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{10} = \frac{\pi}{5}$$

Periodic with fundamental period $T_0 = \frac{\pi}{5}$

(b) $x_2(t) = e^{(-1+j)t}$
 $= e^{-t} e^{jt}$

The e^{-t} term causes the amplitude to continuously decay.

But here $x(t+T) = x(t)$
 $e^{-(t+T)} \neq e^{-t}$

For any $T > 0$

Not Periodic

(c) $x_3[n] = e^{j7\pi n}$

For a discrete-time complex exponential $e^{j\omega n}$

to be periodic, there must exist a positive integer

N such that $\omega N = 2\pi K$

For some integer K

$$\omega = 7\pi$$

$$7\pi N = 2\pi K$$

$$7N = 2K$$

$$N = 2$$

$$\exists (2) = 14 = 2(7)$$

Periodic with $N_0 = 2$

$$(d) \quad x_4[n] = 3e^{j\frac{2\pi}{5}(n+\frac{1}{2})}$$

$$= 3e^{j\frac{2\pi}{5}n} e^{j\frac{2\pi}{10}}$$

The constant phase term does not affect the period
 so we only need to consider $e^{j\frac{2\pi}{5}n}$

$$\frac{2\pi}{5}N = 2\pi K$$

$$3N = 10K$$

The smallest N is

$$N = 10$$

Periodic with
 fundamental period.

$$(e) \quad x_5[n] = 3e^{j\frac{2\pi}{5}n}$$

Again, the n -dependent frequency is

$$\omega_0 = \frac{2\pi}{5}$$

For a discrete-time sinusoid to be periodic
 $\frac{\omega_0}{2\pi}$ must be a rational number

$$\text{Here, } \frac{\omega_0}{2\pi} = \frac{2/5}{2\pi} = \frac{2}{10\pi}$$

since π is irrational, $\frac{2}{10\pi}$ is irrational

Not Periodic.

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$$(a) \quad x(1-t)$$

we need

$$1-t \geq 0$$

therefore

$$-t \leq 2$$

multiply by -1 reverse the inequality

$$t \geq -2$$

then,

$$x(1-t) = 0 \quad \text{for } t \geq -2$$

$$(b) \quad x(1-t) + x(2-t)$$

For the sum to be guaranteed zero, both terms must be individually be guaranteed zero.

First:

$$1-t \leq 0$$

gives

$$t > -2$$

Second:

$$2-t \leq 0$$

gives

$$t > -1$$

Both conditions must hold:

$$t > -2 \quad \text{and} \quad t > -1$$

therefore

$$t > -1$$

$$(c) \quad x(1-t) x(2-t)$$

A Product is guaranteed to be zero if at least one factor is guaranteed to be zero.

$$x(1-t) = 0 \quad \text{for } t > -2$$

and

$$x(2-t) = 0 \quad \text{for } t > -1$$

so the union of these 2 ~~test~~ regions is

$$t > -2$$

$$x(1-t)x(2-t) = 0 \quad \text{for } t > -2$$

$$(d) \quad x(3t)$$

$$3t \leq 0$$

divide by 3

$$t \leq 0$$

$$(e) \quad x(t/3)$$

$$\frac{t}{3} \leq 0$$

$$\underline{t \leq 0}$$

$$x(t/3) = 0 \quad \text{for } t \leq 0$$

3

$$x(t) = 2 \cos(10t + 1) - \sin(4t - 1)$$

The phase shifts $+1$ and -1 does not affect the Period

$$2 \cos(10t + 1)$$

$$\omega_1 = 10$$

$$T_1 = \frac{2\pi}{10} = \frac{\pi}{5}$$

$$-\sin(4t - 1)$$

$$\omega_2 = 4$$

$$T_2 = \frac{2\pi}{4} = \frac{\pi}{2}$$

π

We need the smallest T that is common multiple of both:

$$T = \frac{m\pi}{5} = \frac{n\pi}{2}$$

$$\frac{m}{5} = \frac{n}{2}$$

$$m = 5, n = 2$$

$$\underline{T_0 = \pi}$$

4

$$y[n] = x[n] x[n-2]$$

(a) A system is memoryless if the output at time n depends only on the input at the same time n , i.e. only on $x[n]$

$$y[n] = x[n] x[n-2]$$

The output depends on

$x[n]$ - present input.

$x[n-2]$ - past input

Since it depends on $x[n-2]$, the system has memory

No, the system is not memoryless.

~~Q1~~ is

(b) input $x[n] = A \delta[n]$

$$x[n] = A \delta[n]$$

$$x[n-2] = A \delta[n-2]$$

$$y[n] = x[n] x[n-2]$$

$$y[n] = (A \delta[n]) (A \delta[n-2])$$

$$y[n] = A^2 \delta[n] \delta[n-2]$$

$\delta[n]$ is non-zero only at $n=0$

$\delta[n-2]$ is non-zero only at $n=2$.

Therefore

$$\delta[n] \delta[n-2] = 0$$

$$y[n] = 0$$

For every n

$$y[n] = 0 \quad \forall n$$

That is true for any real or complex value of A .

(c) A system is invertible if different inputs always produce output

$$y[n] = x[n] x[n-2]$$

take

$$x_1[n] = 0$$

$$y_1[n] = 0$$

now take

$$x_2[n] = \delta[n]$$

$$y_2[n] = \delta[n] \delta[n-2] = 0$$

$$x_1[n] \neq x_2[n]$$

but

$$y_1[n] = y_2[n] = 0$$

Thus, the system cannot uniquely recover the input from the output

The system is not invertible.

$$n - n_0 \geq 0$$

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$$y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$$

(a) $x[n] \rightarrow y[n]$

where $y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$

$$z[n] = a x_1[n] + b x_2[n]$$

The output is

$$y_z[n] = \sum_{k=n-n_0}^{n+n_0} z[k]$$

$$= \sum_{k=n-n_0}^{n+n_0} (a x_1[k] + b x_2[k])$$

$$y_z[n] = a \sum_{k=n-n_0}^{n+n_0} x_1[k] + b \sum_{k=n-n_0}^{n+n_0} x_2[k]$$

$$y_z[n] = a y_1[n] + b y_2[n]$$

Hence the system superposition

The system is Linear

(b) For the system to be time-invariant, if $x[n] \rightarrow y[n]$

then a shifted input

$$x_s[n] = x[n-n_1]$$

must produce

$$y_s[n] = y[n-n_1]$$

Let's check

$$y_s[n] = \sum_{k=n-n_0}^{n+n_0} x[k-n_1]$$

$$m = k - n_1$$

$$k = m + n_1$$

$$y_s[n] = \sum_{m=n-n_1-n_0}^{n-n_1+n_0} x[m]$$

$$y[n-n_1] = \sum_{m=(n-n_1)-n_0}^{(n-n_1)+n_0} x[m]$$

which is exactly same

Thus $y_s[n] = y[n-n_1]$ Hence Time-invariant

(c)

$$|x[k]| < B$$

For every n

The output is

$$y[n] = \sum_{k=n-n_0}^{n+n_0} x[k]$$

Step-1: Count the number of terms

The sum goes from $n-n_0$ to $n+n_0$

$$\text{no. of terms} = (n+n_0) - (n-n_0) + 1 = 2n_0 + 1$$

Step-2: Apply the triangle inequality

$$\begin{aligned} |y[n]| &= \left| \sum_{k=n-n_0}^{n+n_0} x[k] \right| \\ &\leq \sum_{k=n-n_0}^{n+n_0} |x[k]| \end{aligned}$$

Since

$$|x[k]| < B,$$

we get

$$|y[n]| < (2n_0 + 1)B$$

$$C = (2n_0 + 1)B$$

Hence the system is BIBO stable.

System is stable because

$$|y[n]| < (2n_0 + 1)B$$

(b)

$$\phi_{xy}(t) = \int_{-\infty}^{\infty} x(t+z) y(z) dz$$

$$(a) \quad \phi_{xy}(t) = \int_{-\infty}^{\infty} y(t+z) x(z) dz$$

$$\phi_{xy}(-t) = \int_{-\infty}^{\infty} x(-t+z) y(z) dz$$

$$\text{let } u = z - t$$

$$z = u + t$$

$$\phi_{xy}(-t) = \int_{-\infty}^{\infty} x(u) y(u+t) du$$

$$\phi_{xy}(-t) = \int_{-\infty}^{\infty} y(t+u) x(u) du$$

$$\boxed{\phi_{xy}(-t) = \phi_{xy}(t)}$$

(b)

$$\phi_{xx}(t) = \int_{-\infty}^{\infty} x(t+z)x(z)dz$$

From part (a), putting $x=y$

$$\phi_{xx}(t) = \phi_{xx}(-t)$$

The odd part of any function $b(t)$ is

$$b_{\text{odd}}(t) = \frac{b(t) - b(-t)}{2}$$

$$b_{\text{xx, odd}}(t) = \frac{\phi_{xx}(t) - \phi_{xx}(-t)}{2}$$

$$\phi_{xx}(t) = \phi_{xx}(-t)$$

we get

$$\boxed{\phi_{\text{xx, odd}}(t) = 0}$$

(c)

Given $y(t) = x(t+\tau)$

$$\phi_{xy}(t) = \int_{-\infty}^{\infty} x(t+z)y(z)dz$$

$$y(z) = x(z+\tau)$$

$$\phi_{xy}(t) = \int_{-\infty}^{\infty} x(t+z)x(z+\tau)dz$$

$$u = z + \tau$$

$$z = u - \tau$$

$$\phi_{xy}(t) = \int_{-\infty}^{\infty} x(t+u-\tau)x(u)du$$

$$\phi_{xy}(t) = \int_{-\infty}^{\infty} x((t-\tau)+u)x(u)du$$

$$\phi_{xx}(s) = \int_{-\infty}^{\infty} x(s+u)x(u)du$$

$$\boxed{\phi_{xy}(t) = \phi_{xx}(t-\tau)}$$

Now find $\phi_{yy}(t)$

$$\phi_{yy}(t) = \int_{-\infty}^{\infty} y(t+z)y(z)dz$$

$$y(t) = x(t+\tau)$$

$$\phi_{yy}(t) = \int_{-\infty}^{\infty} x(t+z+\tau)x(z+\tau)dz$$

$$u = z + \tau$$

$$\phi_{yy}(t) = \int_{-\infty}^{\infty} x(t+u)x(u)du$$

$$\boxed{\phi_{yy}(t) = \phi_{xx}(t)}$$

This makes sense, shifting a signal in time does not change its autocorrelation

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$$y[n] = \frac{1}{4} y[n-1] + x[n]$$

and $x[n] = \delta[n-1]$

we need to determine $y[n]$.

Since the system is causal, assume initial rest:

$$y[n] = 0 \text{ for } n < 0.$$

we can calculate Recursively.

At $n=0$

$$y[0] = \frac{1}{4} y[-1] + \delta[-1]$$

Both terms are zero:

$$y[0] = 0$$

At $n=1$

$$y[1] = \frac{1}{4} y[0] + \delta[0] \\ = 0 + 1 = 1$$

At $n=2$

$$y[2] = \frac{1}{4}$$

At $n=3$

$$y[3] = \frac{1}{4} y[2] = \frac{1}{16}$$

$$y[n] = \frac{1}{4^n}$$

$$y[n] = \begin{cases} 0 & n < 1 \\ \left(\frac{1}{4}\right)^{n-1}, & n \geq 1 \end{cases}$$

$$y[n] = \left(\frac{1}{4}\right)^{n-1} u[n-1]$$

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(a)

$$x[n] = 2^n u[n]$$

$$h[n] = B^n u[n], \quad B \neq 0$$

using convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

Because both signals are right-sided, we need

$$k \geq 0$$

$$n-k \geq 0$$

$$0 \leq k \leq n$$

Therefore

$$y[n] = \sum_{k=0}^n \alpha^k \beta^{n-k}$$

Take β^n outside $y[n] = \beta^n \sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k$

Using the geometric-series formula

$$\sum_{k=0}^n r^k = \frac{1-r^{n+1}}{1-r}$$

$$y[n] = \frac{\beta^n (1 - (\alpha/\beta)^{n+1})}{1 - \alpha/\beta}$$

$$y[n] = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} u[n]$$

$$r[n] + h[n] = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} u[n]$$

(b) $r[n] = h[n] = \alpha^n u[n]$

Since the two signals are identical, we use the Result, use the result from (a), but here $\alpha = \beta$, so we can't directly use that formula.

we have $y[n] = \sum_{k=0}^n \alpha^k \alpha^{n-k}$

Since

$$\alpha^k \cdot \alpha^{n-k} = \alpha^n$$

Since every term is α^n

There are $n+1$ terms.

Therefore

$$\boxed{r[n] + h[n] = (n+1) \alpha^n u[n]}$$

(c) $y[n] = \sum_{k=-\infty}^{\infty} r[k] + h[n-k]$

$$y[n] = \sum_k \left(-\frac{1}{2}\right)^k u[k-4] 4^{n-k} u[2-(n-k)]$$

The first unit step requires $k \geq 4$

The second requires $2-(n-k) \geq 0$

$$k \geq n-2$$

$$k \geq \max(4, n-2)$$

$$y[n] = \sum_{k=\max(4, n-2)}^{\infty} \left(-\frac{1}{2}\right)^k 4^{n-k}$$

Rewrite $y[n] = 4^n \sum_{k=\max(4, n-2)}^{\infty} \left(-\frac{1}{2}\right)^k$

Using

$$\sum_{k=m}^{\infty} r^k = \frac{r^m}{1-r}$$

$$r = -\frac{1}{8}$$

$$y[n] = 4^n \frac{(-\frac{1}{8})^n}{1 - \frac{1}{8}}$$

$$n = \max(4, n-2)$$

$$1 - \frac{1}{8} = \frac{7}{8}$$

$$y[n] = \frac{8}{7} \cdot 4^n \left(-\frac{1}{8}\right)^{\max(4, n-2)}$$

$$y[n] = \begin{cases} \frac{4^n}{4608} & n \leq 6 \\ \frac{8}{7} 4^n \left(-\frac{1}{8}\right)^{n-2} & n \geq 6 \end{cases}$$

(d) For signal shown in figure, both are rectangular, so convolution is just counting number of overlapping samples.

$$y[n] = [1, 2, 3, 4, 5, 5, 4, 3, 2, 2, 2, 3, 4, 5, 5, 4, 3, 2, 1]$$

for $n = 2 \rightarrow 20$

and $y[n] = 0$ otherwise

(a)

$$x[n] = u[n-2] - u[n-6]$$

$$u[n-2] = 1 \quad n \geq 2$$

$$u[n-6] = 1 \quad n \geq 6$$

$$x[n] = \begin{cases} 1 & 2 \leq n \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$$x(e^{j\omega}) = \sum_{n=2}^5 e^{-j\omega n}$$

$$x(e^{j\omega}) = e^{-j2\omega} + e^{-j3\omega} + e^{-j4\omega} + e^{-j5\omega}$$

$$x(e^{j\omega}) = \frac{e^{-j2\omega} [1 - e^{-j4\omega}]}{1 - e^{-j\omega}}$$

$$b) \quad x[n] = \left(\frac{1}{2}\right)^{-n} u[-n-1]$$

The unit step gives

$$-n-1 \geq 0$$

$$n \leq -1$$

$$n \leq -1$$

$$\left(\frac{1}{2}\right)^{-n} = 2^n$$

$$x[n] = 2^n, \quad n \leq -1$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{-1} 2^n e^{-j\omega n}$$

Let

$$m = -n$$

Then $m = 1, 2, 3, \dots$ given

$$X(e^{j\omega}) = \sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m e^{j\omega m}$$

$$r = \frac{1}{2} e^{j\omega}$$

$$X(e^{j\omega}) = \frac{\frac{1}{2} e^{j\omega}}{1 - e^{j\omega}}$$

$$\text{Thus } X(e^{j\omega}) = \frac{e^{j\omega}}{2 - e^{j\omega}}$$

$$c) \quad x[n] = \left(\frac{1}{3}\right)^{|n|} u[-n-2]$$

The unit step gives

$$-n-2 \geq 0$$

$$n \leq -2$$

For $n \leq -2$

$$|n| = -n$$

$$\text{Therefore } x[n] = \left(\frac{1}{3}\right)^{-n} = 3^n$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{-2} 3^n e^{-j\omega n}$$

Let

$$m = -n$$

Then $m = 2, 3, 4, \dots$

$$\text{So } X(e^{j\omega}) = \sum_{m=2}^{\infty} \left(\frac{1}{3}\right)^m e^{j\omega m}$$

$$X(e^{j\omega}) = \frac{\left(\frac{1}{3} e^{j\omega}\right)^2}{1 - \frac{1}{3} e^{j\omega}}$$

$$(d) \quad x(n) = 2^n \sin\left(\frac{\pi n}{4}\right) u(-n)$$

$$u(-n) = 1 \quad \forall n \leq 0$$

$$x(e^{j\omega}) = \sum_{n=-\infty}^0 2^n \sin\left(\frac{\pi n}{4}\right) e^{-j\omega n}, \quad \text{let } m = -n$$

$$\text{Then } x(e^{j\omega}) = - \sum_{m=0}^{\infty} \left(\frac{e^{j\omega}}{2}\right)^m \sin\left(\frac{\pi m}{4}\right)$$

$$\text{Using } \sum_{m=0}^{\infty} r^m \sin(m\theta) = \frac{r \sin \theta}{1 - 2r \cos \theta + r^2}$$

$$\text{with } r = \frac{e^{j\omega}}{2}, \quad \theta = \frac{\pi}{4}$$

$$\text{we get } x(e^{j\omega}) = \frac{-\frac{e^{j\omega}}{2} \cdot \frac{\sqrt{2}}{2}}{1 - \sqrt{2} \frac{e^{j\omega}}{2} + \frac{e^{j2\omega}}{4}}$$

$$\text{So, } x(e^{j\omega}) = \frac{-\sqrt{2} e^{j\omega}}{4 - 2\sqrt{2} e^{j\omega} + e^{j2\omega}}$$

(10)

$$(a) \quad \text{given } x(e^{j\omega}) = \begin{cases} 1 & |\omega| \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right] \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Therefore } x(n) = \frac{1}{2\pi} \left[\int_{-\frac{3\pi}{4}}^{-\frac{\pi}{4}} e^{j\omega n} d\omega + \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} e^{j\omega n} d\omega \right]$$

$$\text{For } n \neq 0 \quad x(n) = \frac{1}{\pi n} \left[\sin\left(\frac{3\pi n}{4}\right) - \sin\left(\frac{\pi n}{4}\right) \right]$$

$$\text{For } n=0, \quad x(n) = \frac{1}{2} \Rightarrow x(n) = \begin{cases} \frac{\sin\left(\frac{3\pi n}{4}\right) - \sin\left(\frac{\pi n}{4}\right)}{\pi n} & n \neq 0 \\ \frac{1}{2} & n=0 \end{cases}$$

$$(b) \quad x(e^{j\omega}) = 1 + 3e^{-j\omega} + 2e^{-j2\omega} - 4e^{-j3\omega} - e^{-j4\omega}$$

$$\text{Using DTFS property } e^{-j\omega k} \leftrightarrow \delta[n-k]$$

we directly obtain

$$x(n) = \delta[n] + 3\delta[n-1] + 2\delta[n-2] - 4\delta[n-3] - \delta[n-4]$$

$$(c) \quad x(e^{j\omega}) = e^{j\omega/2}, \quad \omega \in [-\pi, \pi]$$

Using inverse DTFS

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega/2} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega(n+\frac{1}{2})} d\omega$$

$$\text{Integrating } x(n) = \sin\left[\pi\left(n+\frac{1}{2}\right)\right]$$

$$\text{since } \sin\left(\pi\left(n+\frac{1}{2}\right)\right) = (-1)^n, \quad \text{we get } x(n) = \frac{(-1)^n}{n(n+\frac{1}{2})}$$

(d) $x(e^{j\omega}) = \cos^2 \omega + \sin^2 3\omega$
 Using $\cos^2 \omega = \frac{1 + \cos 2\omega}{2}$

and $\sin^2 3\omega = \frac{1 - \cos 6\omega}{2}$

We get $x(e^{j\omega}) = 1 + \frac{1}{2} \cos(2\omega) - \frac{1}{2} \cos(6\omega)$

Now, $\cos(k\omega) = \frac{1}{2} (e^{jk\omega} + e^{-jk\omega})$

Therefore $x(e^{j\omega}) = 1 + \frac{1}{4} e^{j2\omega} + \frac{1}{4} e^{-j2\omega} - \frac{1}{4} e^{j6\omega} - \frac{1}{4} e^{-j6\omega}$

Using $e^{-j\omega k} \leftrightarrow \delta[n-k] - \frac{1}{4} e^{-j6\omega}$

We get $x[n] = \delta[n] + \frac{1}{4} \delta[n-2] + \frac{\delta[n+2]}{4} - \frac{\delta[n-6]}{4} - \frac{\delta[n+6]}{4}$

(11)

We are given FT as shown in figure.

We know that $x[n] p[n] \leftrightarrow \frac{1}{2\pi} x(e^{j\omega}) P(e^{j\omega})$

(a) $p[n] = \sin\left(\frac{\pi n}{2}\right)$

$p[n] = \sin\left(\frac{\pi n}{2}\right) \leftrightarrow \frac{j}{2} (e^{j\frac{\pi n}{2}} - e^{-j\frac{\pi n}{2}})$

Using DFT Pair

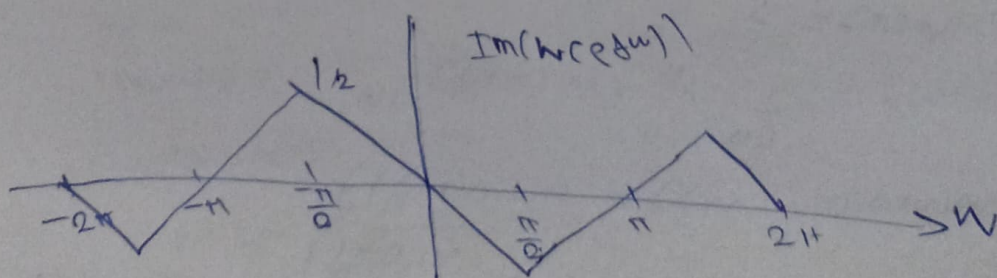
$e^{j\omega n} \leftrightarrow 2\pi \sum_k \delta(\omega - \omega_0 - 2\pi k)$

We have $P(e^{j\omega}) = \frac{\pi}{2} [\delta(\omega - \frac{\pi}{2}) - \delta(\omega + \frac{\pi}{2})]$

Therefore $w(e^{j\omega}) = \frac{1}{2\pi} x(e^{j\omega}) \cdot x P(e^{j\omega})$ which gives

$w(e^{j\omega}) = \frac{1}{2\pi} [x(e^{j(\omega - \frac{\pi}{2})}) - x(e^{j(\omega + \frac{\pi}{2})})]$

$= \frac{1}{2\pi} [x(e^{j(\omega - \frac{\pi}{2})}) - x(e^{j(\omega + \frac{\pi}{2})})]$



(b) Given $p(n) = \sum_{k=-\infty}^{\infty} \delta(n-2k)$

This is an impulse train that is equal to 1 for even n and 0 for odd n .

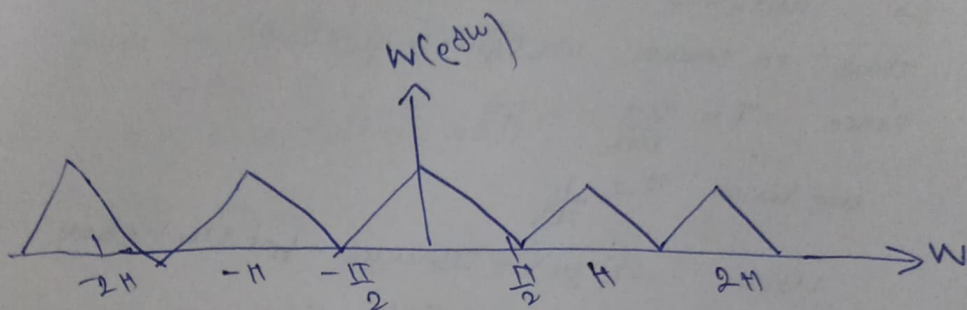
If DFT is $P(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \delta(\omega - k\pi)$

Therefore $w(e^{j\omega}) = \frac{1}{2\pi} X(e^{j\omega}) \cdot P(e^{j\omega})$

Giving $w(e^{j\omega}) = \frac{1}{2} \sum_{k=-\infty}^{\infty} X(e^{j(\omega - k\pi)})$

Because $X(e^{j\omega})$ is 2π periodic, this becomes

$w(e^{j\omega}) = \frac{1}{2} [X(e^{j\omega}) + X(e^{j(\omega - \pi)})]$



12 Given $x(t) = \left(\frac{\sin(50\pi t)}{\pi t} \right)^2$ and sampling freq.

$\omega_s = 150\pi$, we need the max. ω_0 such that

$X(j\omega) = 75 X_1(j\omega)$ for $|\omega| \leq \omega_0$

~~the~~ $\frac{\sin(\omega_c t)}{\pi t} \leftrightarrow \begin{cases} 1 & |\omega| < \omega_c \\ 0 & \text{otherwise} \end{cases}$

here $\omega_c = 50\pi$

Therefore $\frac{\sin(50\pi t)}{\pi t} \leftrightarrow X_1(j\omega)$ where $X_1(j\omega)$ is

rectangular from -50π to 50π since $x(t) = x_1^2(t)$

$X(j\omega) = \frac{1}{2\pi} X_1(j\omega) \cdot X_1(j\omega)$

It also not exist

$$\text{This gives } x_1(j\omega) = \begin{cases} 50 - \frac{j\omega}{2} & |\omega| \leq 100\pi \\ 0 & |\omega| > 100\pi \end{cases}$$

So spectrum extends from -100π to 100π

For sampling with brev. $\omega_s = \omega_c(j\omega)$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} x_c(j\omega - k\omega_s)$$

Since $\omega_s = 150\pi$

The shifted spectra are centered at $0 \pm 150\pi, \pm 300\pi$
original spectrum extends to 100π

The nearest shifted copy centered at 150π
starts at $150\pi - 100\pi = 50\pi$

So $|\omega| \leq 50$

Thus in region $\omega_c(j\omega) = \frac{1}{T} x(j\omega)$

$$\text{Since } T = \frac{2\pi}{\omega_s} = \frac{2\pi}{150\pi} = \frac{1}{75}$$

$$\text{we have } T = \frac{1}{75}$$

Hence $\omega_c(j\omega) = 75 x(j\omega)$ for $|\omega| \leq 50\pi$

Therefore $\omega_0 = 50\pi \text{ rad/s}$

$$\boxed{13} (a) \quad x[n] = \left(-\frac{1}{3}\right)^n u[n-2]$$

$$u[-n-2] = 1 \quad \text{for } n \leq -2$$

$$x[n] = \left(-\frac{1}{3}\right)^n, \quad n \leq -2$$

$$X(z) = \sum_{n=-\infty}^{-2} \left(-\frac{1}{3}\right)^n z^{-n}$$

$$\text{Let } m = -n$$

$$m = 2 \dots \infty$$

$$X(z) = \sum_{m=2}^{\infty} \left(-\frac{1}{3}\right)^{-m} z^m$$

$$\left(-\frac{1}{3}\right)^{-m} = (-3)^m$$

$$X(z) = \sum_{m=2}^{\infty} (-3z)^m$$

$$\sum_{m=2}^{\infty} r^m = \frac{r^2}{1-r}$$

$$X(z) = \frac{(-3z)^2}{1+3z}$$

$$X(z) = \frac{9z^2}{1+3z}$$

ROC

$$|z| < 1$$

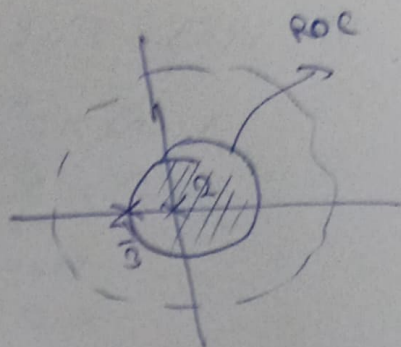
$$|z| < \frac{1}{3}$$

Poles and zeros

$$x(z) = \frac{z^2}{1+z^2}$$

$$1+z^2=0$$

$$z = \pm \frac{j}{3}$$



The DTFT exists if the unit circle

$|z|=1$ lies inside the ROC

But $|z| < \frac{1}{3}$

does not include the unit circle.

FT does not exist

b) $(\frac{1}{4})^n u(3-n)$

$$x(n) = (\frac{1}{4})^n, n \leq 3$$

$$x(z) = \sum_{n=-\infty}^3 (\frac{1}{4})^n z^{-n}$$

$$m = 3-n$$

$$n = 3-m$$

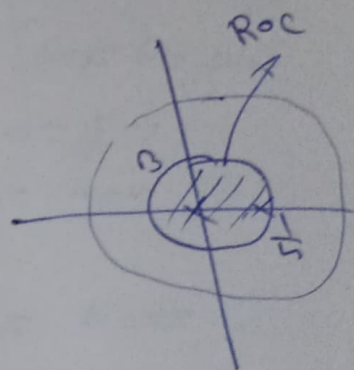
where $m=0, 1, 2, \dots$

$$x(z) = \sum_{m=0}^{\infty} (\frac{1}{4})^{3-m} z^{-(3-m)}$$

$$x(z) = \frac{1}{64} z^{-3} \sum_{m=0}^{\infty} (4z)^m$$

$$x(z) = \frac{1}{64} z^{-3} \left(\frac{1}{1-4z} \right)$$

$$x(z) = \frac{1}{64 z^3 (1-4z)}$$



ROC

$$|4z| < 1$$

$$0 < |z| < \frac{1}{4}$$

Poles and zeros

Poles: $z=0$ (order 3)

$$\text{and } 1-4z=0$$

$$z = \frac{1}{4}$$

There are no branch zeros.

FT does not exist

$$(c) \quad x[n] = 2^n u[n-1] + \left(\frac{1}{4}\right)^n u[n-1]$$

$$x_1[n] = 2^n u[n-1]$$

$$x_1(z) = \sum_{n=-\infty}^{\infty} 2^n z^{-n}$$

$$m = -n$$

$$x_1(z) = \sum_{m=0}^{\infty} \left(\frac{z}{2}\right)^m$$

$$x_1(z) = \frac{1}{1 - z/2}$$

with ROC $|z| < 2$

$$x_2[n] = \left(\frac{1}{4}\right)^n u[n-1]$$

$$x_2(z) = \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n z^{-n}$$

$$x_2(z) = \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n z^{-n}$$

$$x_2(z) = \frac{1/4z}{1 - 1/4z}$$

$$x_2(z) = \frac{1}{4z - 1}$$

with ROC $|z| > \frac{1}{4}$

combine

$$x(z) = \frac{1}{1 - z/2} + \frac{1}{4z - 1}$$

$$x(z) = \frac{7z}{2(1 - z/2)(4z - 1)}$$

ROC $|z| < 2$ $|z| > \frac{1}{4}$

$$\frac{1}{4} < |z| < 2$$

Poles and zeros

$$1 - \frac{z}{2} = 0 \quad \boxed{z = 2}$$

$$4z - 1 = 0 \quad \boxed{z = \frac{1}{4}}$$

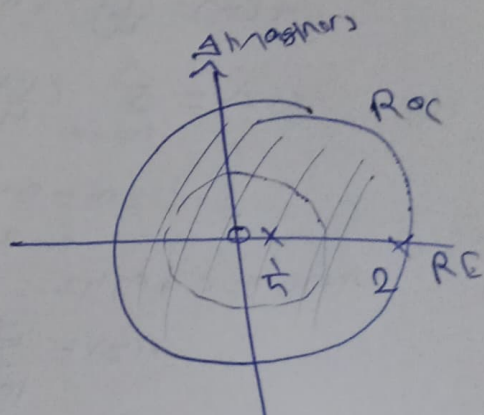
zero at $z = 0$

$$\frac{1}{4} < 1 < 2$$

Therefore, the unit circle

lies inside ROC

hence FT exists



$$(d) \quad x[n] = \left(\frac{1}{3}\right)^{n-2} u[n-2]$$

$$u[n-2] = 1 \quad \text{for } n \geq 2$$

$$x(z) = \sum_{n=2}^{\infty} \left(\frac{1}{3}\right)^{n-2} z^{-n}$$

$$m = n - 2$$

$$n = m + 2$$

$$x(z) = \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^m z^{-(m+2)}$$

$$x(z) = z^{-2} \sum_{m=0}^{\infty} \left(\frac{1}{3z}\right)^m$$

$$x(z) = \frac{1}{z^2} \left(\frac{3z}{3z-1} \right)$$

$$x(z) = \frac{3}{z(3z-1)}$$

$$\text{ROC} \quad \left| \frac{1}{3z} \right| < 1$$

$$\text{we need } |z| > \frac{1}{3}$$

Poles and
zero)

$$x(z) = \frac{3}{z(3z-1)}$$

$$\boxed{z=0}$$

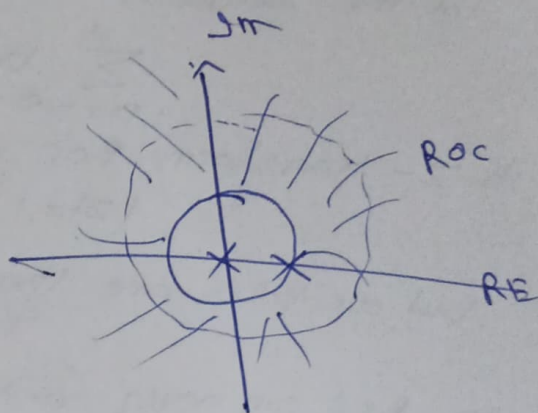
$$\text{and } 3z-1=0$$

$$z = \frac{1}{3}$$

There are not finite
zero

$$|z| > \frac{1}{3}$$

FT exist.



111

(a)

$$X(z) = \frac{1 - 2z^{-1}}{1 + \frac{1}{2}z^{-1} + z^{-2}}$$

Let

$$r = z^{-1}$$

Then the

denominator is $\rightarrow 1 + \frac{1}{2}r + r^2$

$$\text{roots } r^2 + \frac{1}{2}r + 1 = 0$$

$$r = -\frac{1}{4} \pm j\frac{\sqrt{15}}{4}$$

$$|r| = 1$$

In fact the poles are $z = -\frac{1}{4} \pm j\frac{\sqrt{15}}{4}$ and both lie exactly on the unit circle.

Step-2

check the absolute - summability condition.

For the sequence to be absolutely summable

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty,$$

its Z-transform ROC must include the unit circle.

$$|z| = 1$$

But our poles are themselves on

$$|z| = 1$$

An ROC can never contain a pole

Therefore, there is no possible ROC that includes the unit circle.

The possible ROCs are only

$$|z| > 1$$

or

$$|z| < 1$$

depending on whether the seq. is right-sided or left-sided.

Neither includes the unit circle.

No absolutely summable seq. corresponds to the given $X(z)$.

14 (b) Long Division

$$x(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{1}{2}z^{-1}}$$

$x(z)$ is Right-sided

Since the seq. is right-sided, the ROC is outside the outermost pole.

The pole is obtained from

$$1 + \frac{1}{2}z^{-1} = 0$$

$$z = -\frac{1}{2}$$

$$\boxed{\text{ROC: } |z| > \frac{1}{2}}$$

$$x(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{1}{2}z^{-1}}$$

$$1 - \frac{1}{2}z^{-1} = -(1 + \frac{1}{2}z^{-1}) + 2$$

$$x(z) = -1 + \frac{2}{1 + \frac{1}{2}z^{-1}}$$

$$\frac{1}{1 - az^{-1}} \leftrightarrow a^n u(n)$$

$$\frac{1}{1 + \frac{1}{2}z^{-1}} \leftrightarrow (-\frac{1}{2})^n u(n)$$

$$x(n) = -\delta(n) + 2(-\frac{1}{2})^n u(n)$$

$$x(n) = -\delta(n) + 2(-\frac{1}{2})^n u(n)$$

Let verify the first few samples

At $n=0$

$$x(0) = -1 + 2 = 1$$

At $n=1$

$$x(1) = 2(-\frac{1}{2}) = -1$$

At $n=2$

$$x(2) = 2(\frac{1}{4}) = \frac{1}{2}$$

At $n=3$

$$x(3) = 2(-\frac{1}{8}) = -\frac{1}{4}$$

so the seq. is

$$x(n) = \begin{bmatrix} 1 & n=0 \\ -1 & n=1 \\ \frac{1}{2} & n=2 \\ -\frac{1}{4} & n=3 \\ \frac{1}{8} & n=4 \\ \vdots & \vdots \end{bmatrix}$$

15 (a) $w(z) = \frac{z-1/2}{z^2 + \frac{1}{2}z - \frac{3}{16}}$

Poles

$$(z - \frac{1}{4})(z + \frac{3}{4})$$

$$z = \frac{1}{4}, -\frac{3}{4}$$

$$|\frac{1}{4}| = \frac{1}{4}$$

$$|-\frac{3}{4}| = \frac{3}{4}$$

$$\angle |z| = \frac{3}{4}$$

outermost

Poles

step-2 Use stability

$$|z| = 1$$

$$|z| > \frac{3}{4}$$

$$|z| < \frac{1}{4}$$

$$\frac{1}{4} < |z| < \frac{3}{4}$$

$$|z| > \frac{3}{4}$$

Containing the unity circle.

Step-1

For a causal system, the ROC must be outside the outermost pole

$$|z| > \frac{3}{4}$$

system (a) is causal.

(15) (b) $H(z) = \frac{z+1}{z + \frac{4}{3} - \frac{1}{2}z^{-2} - \frac{2}{3}z^{-3}}$

$$H(z) = \frac{z^3(z+1)}{z^4 + \frac{4}{3}z^3 - \frac{1}{2}z - \frac{2}{3}}$$

$$\frac{1}{6}(3z+4)(2z^3-1)$$

$$3z+4=0$$

$$z = -\frac{4}{3}$$

$$2z^3-1=0$$

$$z^3 = \frac{1}{2}$$

$$|z| = \sqrt[3]{\frac{1}{2}}$$

So the pole magnitude are

$$\frac{4}{3}, \sqrt[3]{\frac{1}{2}}$$

with $\frac{4}{3} > \sqrt[3]{\frac{1}{2}}$

Step-2

(16) Determine the stable ROC

$$|z| > 0.794$$

$$|z| = \frac{4}{3} \approx 1.33$$

the ROC that contains the unit circle is

$$\sqrt[3]{\frac{1}{2}} < |z| < \frac{4}{3}$$

$$0.794 < 1 < 1.33$$

check-1

For a causal system, the ROC must be

$|z| > \text{outermost pole}$.

$$|z| > \frac{4}{3}$$

But this ROC does not contain the unit circle, so it would not be stable. stable ROC is instead an annulus.

$$(\frac{1}{2})^{\frac{1}{3}} < |z| < \frac{4}{3}$$

An annular ROC corresponds to a two-sided impulse response, not a causal one.